Your Paper Title: Subtitle if Needed

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***Abstract*—Deep-learning-based landslide monitoring has shown strong potential for automated mapping from remote sensing imagery, yet its practical deployment remains limited by the cost of pixel-level annotation. Active learning (AL) offers a natural remedy by prioritizing informative samples for labeling; however, existing AL strategies for semantic segmentation typically rely on a single criterion, such as uncertainty or diversity, and therefore adapt poorly to changing learning stages. They also offer limited interpretability for why specific samples are queried. To address these limitations, this paper proposes AAL-SD, an LLM-agent-driven adaptive active learning framework for landslide semantic segmentation. Its core acquisition module, AD-KUCS, combines uncertainty and knowledge gain through an adaptive weighting mechanism governed by a warmup- and risk-aware closed-loop policy. A Cluster-Balanced K' algorithm corrects the dominant-cluster capture bias in knowledge-based sampling by applying saturation-weighted corrections. The LLM agent inspects candidates in context and finalizes the queried subset with an interpretable rationale. On Landslide4Sense, a 15-round evaluation with three independent seeds and seven strategies shows that the proposed framework achieves competitive performance (final mIoU 0.703 for the agent-controlled variant) against strong baselines including Core-Set (0.710) and Entropy (0.707), while uniquely providing auditable natural language justifications. An unexpected observation is that lambda sensitivity remains within 2% across all conditions, suggesting that under severe class imbalance, the knowledge-uncertainty balance is a safe rather than sensitive control variable.**

***Index Terms*—Active learning, landslide detection, semantic segmentation, LLM agent, remote sensing, deep learning.**

I. Introduction

A. Background and Motivation

LANDSLIDES represent one of the most destructive geological hazards globally, causing significant casualties, property damage, and economic losses annually. With the advancement of remote sensing technology and the increasing availability of high-resolution satellite imagery, automated landslide detection has emerged as a critical tool for disaster mitigation and early warning systems [**?**]. Deep learning approaches, particularly semantic segmentation models such as U-Net and DeepLabV3+, have achieved remarkable success in extracting landslide boundaries from optical remote sensing images [**?**], [**?**], [1]. These data-driven methods can automatically learn hierarchical features from raw imagery without manual feature engineering, substantially outperforming traditional machine learning approaches.

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However, the exceptional performance of deep learning models comes at a significant cost: these models are inherently data-hungry and require large-scale annotated datasets to generalize well. In the context of landslide detection, acquiring pixel-level annotations presents unique challenges. First, landslide annotation demands specialized geological expertise, making the labeling process expensive and time-consuming. Second, landslide events exhibit high spatial and temporal variability, with diverse morphological characteristics across different geographical regions, requiring diverse training samples. Third, the class imbalance problem is particularly severe in landslide datasets, as landslide pixels typically constitute only a small fraction of the total image area. These challenges have become the primary bottleneck for the practical deployment of deep learning-based landslide detection systems.

B. Active Learning as a Solution

Active learning (AL) has emerged as a powerful paradigm to address the data annotation bottleneck in machine learning. The core hypothesis of AL is that if learning algorithms can actively select the most informative samples for annotation, they can achieve superior performance with fewer labeled examples compared to passive random sampling. By iteratively selecting valuable samples from an unlabeled pool and querying their labels from an oracle (e.g., human annotators), AL aims to maximize the learning efficiency under a fixed annotation budget.

In the context of semantic segmentation for remote sensing, AL becomes particularly valuable because pixel-level annotation is extremely labor-intensive compared to image-level classification. The annotation process requires experts to delineate precise boundaries of landslide areas, which can take hours per image. Therefore, AL offers substantial potential to reduce the annotation burden while maintaining detection accuracy.

Traditional AL strategies for semantic segmentation can be broadly categorized into three groups: uncertainty-based methods, diversity-based methods, and hybrid methods. Uncertainty-based methods (e.g., entropy sampling, margin sampling) select samples where the model exhibits high prediction uncertainty, aiming to refine decision boundaries. Diversity-based methods (e.g., Core-Set) select samples that maximize coverage of the feature space, promoting exploration. Hybrid methods attempt to combine both perspectives, often using fixed or manually tuned weights.

Despite progress, existing AL strategies suffer from fundamental limitations that hinder their effectiveness in practical landslide detection scenarios. First, single-metric approaches cannot dynamically adapt to the learning progress: early-stage learning may benefit more from exploration, while later

stages should focus on exploitation. Second, these methods lack interpretability; the selection rationale remains opaque to human operators. Third, most strategies rely on fixed weighting schemes without adaptive mechanisms to adjust the exploration-exploitation balance based on model performance feedback.

C. LLM Agents: A New Paradigm

The recent emergence of Large Language Models (LLMs), particularly those with Chain-of-Thought (CoT) reasoning capabilities, has opened new possibilities for intelligent decision-making in machine learning systems. Unlike traditional algorithmic approaches, LLM Agents can perform complex reasoning, understand contextual information, and generate human-interpretable explanations for their decisions. The ReAct framework has demonstrated that LLMs can synergize reasoning and acting by interleaving thought processes with tool invocations and environmental observations.

Inspired by these advances, we propose integrating LLM Agents into the active learning decision loop for landslide semantic segmentation. The agent can receive comprehensive information about the learning state (e.g., uncertainty scores, knowledge gain metrics, training progress) and make informed sample selection decisions through natural language reasoning. This integration yields a cognition-assisted, computation-grounded AL loop, improving contextual decision-making and interpretability while keeping key control variables governed by explicit policies in the main full-model setting.

D. Contributions

The main contributions of this paper are fivefold:

1) **Novel Framework**: We propose AAL-SD, a dual-line control framework that integrates an LLM agent into the active learning decision loop via auditable tool calls. In the main full-model setting, λ_t is provided by a warmup- and risk-aware closed-loop policy, while the agent performs context-aware candidate inspection, final subset selection, and rationale generation.
2) **Adaptive Algorithm with Cluster-Balanced Sampling**: We design AD-KUCS, a dynamic hybrid query strategy that fuses uncertainty and knowledge gain with adaptive weighting. To address the dominant-cluster capture problem inherent in knowledge-based metrics, we introduce Cluster-Balanced $K'(x)$, which orthogonalizes the knowledge-gain direction from the uncertainty direction through saturation-weighted corrections.
3) **Closed-Loop Optimization View**: We interpret AAL-SD as a closed-loop optimization controller rather than a static ranking rule. By regulating query weights and selected samples using gradient-derived risk signals, the framework indirectly reshapes the labeled-data distribution and the downstream gradient trajectory.
4) **Interpretable Decisions**: The LLM Agent provides natural language explanations for each sample selection decision, offering unprecedented transparency in the active learning process. This interpretability is crucial for building trust in automated annotation systems for safety-critical remote sensing applications.
5) **Comprehensive Evaluation**: We conduct extensive experiments on the Landslide4Sense benchmark dataset with 15 active learning rounds, three independent seeds, and seven strategies. We demonstrate that AAL-SD is competitive against strong baselines while providing interpretable control. We additionally report a consistent observation that lambda sensitivity remains within 2%, suggesting that the knowledge-uncertainty balance is a safe rather than sensitive parameter under severe class imbalance.

The remainder of this paper is organized as follows. Section II reviews related work on active learning for semantic segmentation, landslide detection with deep learning, and LLM Agents in machine learning. Section III presents the methodology, including the AD-KUCS algorithm and the LLM Agent decision system. Section IV describes the experimental setup and reports the results with detailed analysis. Section V discusses the implications, limitations, and future work. Section VI concludes the paper.

II. Related Work

A. Active Learning for Semantic Segmentation

Active learning for semantic segmentation has attracted significant research attention in recent years. The goal is to develop query strategies that can effectively identify the most valuable image patches or superpixels for annotation. Existing methods can be broadly classified into three categories.

Uncertainty-based Methods. These approaches select samples where the model predictions are most uncertain. Entropy sampling computes the average pixel-wise entropy across the entire image, selecting samples with the highest prediction entropy. Monte Carlo (MC) Dropout-based methods estimate uncertainty by measuring the variance of predictions across multiple forward passes with dropout enabled. Confidence-based margin sampling selects samples where the difference between the top two predicted class probabilities is minimal. Recent work has demonstrated that uncertainty sampling can achieve comparable performance to full supervision with significantly fewer labeled samples in remote sensing segmentation tasks.

Diversity-based Methods. These approaches aim to select samples that maximize diversity within the selected batch. Core-Set selection formulates the problem as maximizing coverage in the feature space, selecting samples that are most representative of the unlabeled pool. Density-weighted methods consider both uncertainty and the density of samples in the feature space to avoid selecting redundant samples. ViewAL introduced viewpoint entropy for 3D scene understanding, showing that incorporating viewpoint diversity improves active learning performance.

Hybrid Methods. Combining uncertainty and diversity has shown promising results. DIAL (Deep Interactive and Active Learning) proposed interactive learning schemes that integrate user inputs into deep neural networks, along with active learning strategies to guide users toward the most relevant areas to

annotate. The method compared different acquisition functions including ConfidNet, entropy, and ODIN, demonstrating that uncertainty-based active learning can quickly lead users toward model errors. However, these hybrid methods typically use fixed weighting schemes without adaptive mechanisms.

B. Landslide Detection with Deep Learning

Deep learning has revolutionized landslide detection from remote sensing imagery. Semantic segmentation models such as U-Net, SegNet, and DeepLabV3+ have become standard approaches for pixel-wise landslide mapping [**?**], [**?**], [1]. The DeepLabV3+ model, with its atrous separable convolution and encoder-decoder structure, has shown particular effectiveness in capturing multi-scale contextual information for landslide boundary delineation.

The Landslide4Sense dataset, introduced by Ghorbanzadeh et al. [2], provides a standardized benchmark for landslide detection research. This dataset contains multi-spectral Sentinel-2 imagery with pixel-level annotations, enabling fair comparison across different methods. Recent studies have explored various improvements including attention mechanisms, transfer learning, and multi-modal data fusion to enhance landslide detection performance.

However, a critical challenge remains: these methods require substantial labeled data for training. Annotation of landslide imagery demands expert geological knowledge and significant manual effort. The class imbalance problem is particularly severe, as landslide pixels often represent less than 10% of the total image area. These factors motivate the application of active learning to reduce annotation requirements while maintaining detection accuracy.

C. Large Language Models in Machine Learning

The emergence of Large Language Models has transformed various aspects of machine learning. Models with Chain-of-Thought (CoT) prompting capabilities demonstrate remarkable abilities in natural language understanding, reasoning, and code generation. The ReAct framework demonstrates that LLMs can synergize reasoning and acting by interleaving thought processes with tool invocations and environmental observations. This approach enables agents to maintain situational awareness, update their understanding based on observations, and plan future actions.

In the context of data labeling and active learning, prior work has explored using LLMs as annotators. MEGAnto+ introduced a human-LLM collaborative annotation paradigm where LLMs create labels that annotators selectively verify according to active learning strategies. However, these approaches primarily focus on text or image classification tasks, with limited exploration in semantic segmentation and remote sensing applications.

Our work advances this direction by integrating LLM Agents directly into the active learning decision loop for semantic segmentation, enabling dynamic, interpretable sample selection guided by comprehensive contextual reasoning.

Fig. 1: Dual-line architecture of AAL-SD. The Active Learning Line (top, blue) carries the data flow from pool initialization through training, inference, AD-KUCS scoring, and LLM agent selection to pool update. The Learning Control Line (bottom, red) carries gradient diagnostics through risk assessment to the λ policy engine, whose output feeds back into both the scoring block and the agent context.

III. Methodology

A. Problem Formulation

We formalize the active learning problem for landslide semantic segmentation as follows. Let $U = \{x_1, x_2, \ldots, x_n\}$ denote the pool of unlabeled remote sensing images, L_0 a small initial labeled set, and T a fixed test set. At each round $t \in \{1, \ldots, T_{\max}\}$, a segmentation model f_θ is trained on the current labeled set L_t. An acquisition function $a(x, f_\theta)$ scores each unlabeled sample, and the top-k candidates are queried for oracle annotation, forming the query set Q_t. The labeled set is updated as $L_{t+1} = L_t \cup Q_t$, and the process repeats until the annotation budget B is exhausted.

A key perspective motivates this work: the query set Q_t does not merely add labels but reshapes the empirical training distribution $\hat{P}_{t+1}$. Consequently, the optimization objective $\mathcal{L}_{t+1}(\theta)$ changes, and all subsequent gradient trajectories $\nabla_\theta \mathcal{L}_{t+1}$ are indirectly determined by the acquisition decision. This data-selection-as-distribution-control perspective motivates treating the acquisition policy as a feedback controller rather than a static ranking rule.

B. AAL-SD Framework Overview

The AAL-SD framework organizes the pipeline around two interacting loops, illustrated in Fig. 1.

The **Active Learning Line** carries the data flow: the data pools (L_0, U, T) are initialized; DeepLabV3+ [1] is trained on L_t for a fixed number of epochs per round; the trained model performs inference on U to produce pixel-wise probability maps $P(x)$ and deep feature embeddings $f(x)$; the AD-KUCS module computes acquisition scores; the LLM agent inspects candidates and finalizes the queried subset Q_t with an interpretable rationale; and the pool is updated for the next round.

The **Learning Control Line** carries the control flow: per-epoch gradient diagnostics are computed during training; these statistics are aggregated into a risk score and rollback flag; the risk signals feed into the λ policy engine, which determines the acquisition weight λ_t; and the policy-determined λ_t is injected into both the AD-KUCS scoring formula and the agent's system-status context.

The two lines converge at the scoring and agent blocks. This dual-line design ensures that acquisition decisions are not driven by a fixed heuristic but are continuously adapted based on optimization diagnostics, while the agent's interpretable reasoning operates within bounds set by the deterministic control policy.

C. AD-KUCS: Adaptive Dynamic Knowledge-Uncertainty Sampling

The core acquisition module, AD-KUCS, fuses two complementary sample quality signals through an adaptive weighting mechanism.

1) Uncertainty Metric $U(x)$: For a sample x with N pixels and C semantic classes, the model outputs a pixel-wise probability distribution $p_i \in \Delta^{C-1}$ for each pixel i. The sample-level uncertainty is the mean pixel entropy:

$$U(x) = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} p_i(c)\log p_i(c). \tag{1}$$

High entropy indicates that the model assigns diffuse probability mass across classes, marking samples where annotation would most directly refine decision boundaries.

2) Knowledge Gain Metric $K(x)$: Uncertainty sampling alone tends to concentrate selections near existing decision boundaries, potentially neglecting under-represented terrain types in the feature space. To address this, we design a *clustering-based representativeness score* that quantifies how well a candidate typifies an unexplored feature-space region.

Let $\{f_x\}_{x\in U}$ denote the deep feature embeddings of all unlabeled samples, extracted from the penultimate backbone layer via global average pooling. We apply k-means clustering to this embedding set, partitioning U into C_k clusters with centroids $\{\mu_c\}_{c=1}^{C_k}$, where $C_k = \min(k_{\max}, |U|)$. The knowledge gain score for sample x is defined as:

$$K(x) = 1 - \frac{\|f_x - \mu_{c(x)}\|_2}{d_{\max}}, \quad d_{\max} = \max_{x'\in U}\|f_{x'} - \mu_{c(x')}\|_2, \tag{2}$$

where $c(x)$ denotes the cluster assignment of x and $d_{\max}$ normalizes by the largest intra-cluster distance observed across the entire unlabeled pool. A sample with $K(x)$ close to 1.0 lies near its cluster centroid, making it a prototypical representative of a coherent feature-space region. A sample with $K(x)$ near 0.0 is the farthest outlier within its cluster. By favoring high-K samples, the acquisition preferentially selects candidates that are most representative of under-explored terrain clusters, maximizing labeled-set coverage through information-dense annotations rather than isolated outliers.

Unlike Core-Set-style metrics [3], which maximize the minimum distance to the existing labeled set and may therefore favor boundary outliers, the clustering-based $K(x)$ identifies the most central member within each cluster of the unlabeled pool. This yields a more stable coverage signal, as cluster centroids are robust to individual noisy samples in the feature space.

3) Cluster-Balanced Knowledge Gain $K'(x)$: While the raw $K(x)$ metric successfully identifies representative samples within each cluster, it does not account for the evolving saturation of individual clusters as labeling progresses. In practice, we observe a *dominant-cluster capture* phenomenon: a small number of large clusters accumulate the majority of labeled samples across rounds, yet the raw $K(x)$ metric continues to direct selections toward these already-saturated regions. In our landslide segmentation experiments, one cluster (Cluster 19) holds 52% of all labeled samples while $K(x)$ keeps selecting its members, providing diminishing informational returns.

To correct this bias, we introduce Cluster-Balanced $K'(x)$, which re-weights each sample's knowledge score by the inverse saturation of its cluster. The procedure consists of three steps.

Step 1: Saturation computation with Laplace smoothing. For each cluster c, we compute:

$$\mathrm{sat}(c) = \frac{N_c^{\text{labeled}} + \epsilon}{N_c^{\text{pool}} + \epsilon}, \quad \epsilon = 1, \tag{3}$$

where N_c^{labeled} and N_c^{pool} denote the number of labeled and total pool samples in cluster c, respectively. The Laplace smoothing term $\epsilon = 1$ prevents numerical instability for small clusters while having negligible effect on large ones.

Step 2: Cluster weight with noise filtering and truncation. Each cluster receives a weight based on its available capacity:

$$w(c) = \begin{cases} 0 & \text{if } |c| < n_{\min}, \\ \max\big((1-\mathrm{sat}(c))^{\gamma},\ r_{\text{floor}}\cdot w_{\max}\big) & \text{otherwise}, \end{cases} \tag{4}$$

where $n_{\min}$ is the minimum cluster size threshold (filtering noise clusters), $\gamma = 1$ controls saturation penalty strength, $r_{\text{floor}} = 0.1$ prevents any valid cluster from being completely suppressed, and $w_{\max} = \max_c w(c)$. The weights are then normalized to sum to one.

Step 3: Balanced knowledge gain score. The final corrected score multiplies the raw knowledge gain by the cluster weight:

$$K'(x) = K(x)\cdot w\big(\mathrm{cluster}(x)\big). \tag{5}$$

By down-weighting saturated clusters and preserving weight for under-explored regions, $K'(x)$ effectively orthogonalizes the knowledge signal from the uncertainty signal: K' no longer selects from the same high-density regions that entropy-based $U(x)$ already covers. This orthogonalization is empirically verified in Section IV.

4) Acquisition Function: The final acquisition score combines both signals through the adaptive weight $\lambda_t \in [0,1]$:

$$\mathrm{Score}(x) = (1-\lambda_t)\cdot U(x) + \lambda_t \cdot K'(x). \tag{6}$$

Setting $\lambda_t = 0$ reduces the acquisition to pure uncertainty sampling; $\lambda_t = 1$ reduces it to pure cluster-balanced diversity sampling. The weight λ_t is not fixed but is dynamically determined each round by the three-phase adaptive policy described in Section III-D.

D. Three-Phase Adaptive Lambda Policy

The adaptive weight λ_t follows a three-phase policy aligned with the lifecycle of the annotation budget. Each phase corresponds to a distinct operational regime where the relative value of uncertainty versus coverage shifts based on labeled-set size and optimization health.

1) Phase 1: Uncertainty Warmup ($t \leq t_{\text{warm}}$): During the earliest rounds, L_t is very small relative to U. In this regime, the knowledge gain metric $K(x)$ is unreliable: the embedding space has not yet been sufficiently structured by training, and cluster centroids estimated from a small labeled set carry high variance. We therefore enforce $\lambda_t = 0$ during the warmup phase, reducing acquisition to pure uncertainty sampling. This prioritizes decision boundary refinement at the stage where the model is most plastic and most in need of targeted correction.

2) Phase 2: Risk-Aware Modulation ($t_{\text{warm}} < t \leq t_{\text{late}}$): As L_t grows, the embedding space stabilizes and $K(x)$ becomes a reliable coverage signal. In this phase, λ_t is determined by a risk-aware feedback mechanism based on gradient diagnostics. To prevent policy oscillation from round-to-round noise in the raw risk signal, the effective risk is smoothed via exponential moving average:

$$\tilde{r}_t = \beta \cdot \tilde{r}_{t-1} + (1 - \beta) \cdot r_t, \tag{7}$$

where r_t is the raw risk score at round t and β is the smoothing factor. The policy clamps λ_t downward when $\tilde{r}_t$ exceeds a risk threshold τ_{risk}, shifting acquisition toward uncertainty-focused selection under optimization instability. A cooling period of n_{cool} rounds prevents successive adjustments from compounding.

3) Phase 3: Late-Stage Coverage Ramp ($t > t_{\text{late}}$): Near the end of the annotation budget, the marginal value of coverage increases relative to uncertainty refinement. With few labeling rounds remaining, the model is unlikely to undergo large structural changes, so adding further boundary-adjacent samples yields diminishing returns. Conversely, ensuring broad feature-space coverage before annotation ends maximizes the final model's ability to generalize to unseen terrain types. To reflect this shift, λ_t is ramped toward $\lambda_{\max}$ in the terminal rounds:

$$\lambda_t = \lambda_{\max} \cdot \frac{t - t_{\text{late}}}{T_{\max} - t_{\text{late}}}, \tag{8}$$

where $\lambda_{\max} = 0.65$. The cap below 1.0 preserves a diversity floor: the controller never fully abandons uncertainty signals even at budget exhaustion.

E. Gradient-Derived Risk Signals

A central element of the Learning Control Line is the train-validation gradient cosine (TVC), which serves as the primary feedback signal for Phase 2 of the adaptive policy.

At each training epoch, we compute the gradient of the segmentation loss with respect to the model parameters θ on both the current training batch and a held-out validation batch. The TVC for round t is:

$$\text{TVC}_t = \frac{\nabla_\theta \mathcal{L}_{\text{train}} \cdot \nabla_\theta \mathcal{L}_{\text{val}}}{\|\nabla_\theta \mathcal{L}_{\text{train}}\|_2 \, \|\nabla_\theta \mathcal{L}_{\text{val}}\|_2}. \tag{9}$$

When $\text{TVC}_t > 0$, the optimization directions on training and validation data are aligned, indicating consistent generalization. When $\text{TVC}_t < 0$, the gradients are anti-aligned: the update that reduces training loss actively increases validation loss. This gradient interference pattern is a principled indicator of distributional overfitting, with connections to gradient-based interference analyses in continual learning [4].

By reducing λ_t in response to negative TVC, the controller shifts acquisition toward high-uncertainty samples. These samples lie near the current decision boundary and tend to be drawn from regions where the model's predictions are inconsistent across the training and validation distributions. Adding them to L_{t+1} exposes the model to its most uncertain predictions in the subsequent training round, promoting gradient re-alignment without directly modifying the optimizer or loss function. The risk score r_t used in Eq. (7) is derived from TVC_t together with recent performance trend statistics, and is normalized to $[0, 1]$ via adaptive thresholds derived from a rolling window of recent metric variability.

F. LLM Agent Decision System

The LLM agent serves as the interpretable finalization layer within each active learning round, operating after AD-KUCS scoring and before the pool update.

1) Agent Architecture: The agent is implemented as a ReAct loop [5] that interleaves chain-of-thought reasoning with structured tool calls. At each round, the agent receives: (1) current learning statistics (round number, mIoU, budget status, λ_t, and risk indicators); (2) the top-k candidate samples with their $U(x)$, $K(x)$, and $\text{Score}(x)$ values; and (3) historical selection patterns. Using this context, the agent performs multi-step reasoning to inspect candidates, assess spatial and semantic diversity, and produce a final selection with a natural-language rationale.

In the primary full-model configuration (Variant A), the agent does not have access to `set_lambda()`: λ_t is determined exclusively by the closed-loop policy described in Section III-D. This separation ensures that the control logic remains deterministic and auditable, while the agent focuses on contextual candidate inspection. A second variant (Variant B) grants the agent explicit λ-setting authority under constraints, and is evaluated as an ablation in Section IV.

2) Selection Guardrails: To ensure that agent decisions remain within operationally safe bounds, we impose hard constraints applied after each agent selection. Specifically: (1) the median uncertainty of the selected subset must exceed a minimum threshold τ_U; (2) the fraction of low-uncertainty selections is capped at a maximum allowable ratio; and (3) under high-risk conditions ($\tilde{r}_t > \tau_{\text{high}}$), selections are constrained to the high-uncertainty stratum of the candidate pool. These guardrails define a safe operating region within which the agent exercises contextual judgment. If a selection violates any constraint, the system reverts to score-ranked selection for that round and logs a safety override event, preserving full auditability.

G. AAL-SD as a Discrete-Time Feedback Controller

The preceding components form a coherent discrete-time feedback control system. At each round t, the system state is $s_t = (t, \lambda_t, \text{mIoU}_t, \tilde{r}_t)$. The control variable is λ_{t+1}, determined by the three-phase policy as a function of s_t. The feedback signal is TVC_t, which measures gradient alignment

TABLE I: Dataset Configuration and Active Learning Protocol

Parameter	Value
Total training pool	3,039 samples
Initial labeled set L_0	151 samples (5%)
Unlabeled pool U	2,888 samples
Fixed test set T	760 samples
Per-round query budget	88 samples
Number of AL rounds	15
Final labeled set size	1,471 samples
Independent seeds	3 (seeds 42, 43, 44)

TABLE II: Final Performance Comparison on Landslide4Sense (15-Round, 3-Seed Mean $\pm$ Std)

Strategy	Type	Final mIoU	Range
Core-Set	Diversity	$0.7101 \pm X$	0.17%
BALD	Uncertainty	$0.7080 \pm X$	2.52%
Entropy	Uncertainty	$0.7065 \pm X$	0.71%
Model B	Adaptive	$0.7027 \pm X$	0.76%
Model A	Adaptive	$0.6971 \pm X$	1.99%
Random	None	$0.6945 \pm X$	2.95%
DIAL	Hybrid	$0.6932 \pm X$	3.27%

between the training and validation distributions. The effect of the control action propagates through data selection to the labeled distribution $\hat{P}_{t+1}$ and then to the downstream gradient trajectory $\nabla_\theta \mathcal{L}_{t+1}$.

We do not claim analytical stability guarantees, as the dynamics of neural network training under distribution shift are not tractable in closed form. Instead, the empirical gradient-coupling analysis in Section IV provides evidence that the feedback loop responds consistently to optimization diagnostics: rounds with positive TVC receive higher λ_t (stronger coverage emphasis), while rounds with negative TVC receive lower λ_t (stronger uncertainty emphasis). This consistent co-variation supports the interpretation that AAL-SD indirectly shapes optimization trajectories through data selection, rather than through direct modification of the loss or optimizer.

IV. Experiments

We evaluate the proposed AAL-SD framework on the Landslide4Sense benchmark through a comprehensive 15-round active learning protocol with three independent seeds. This section presents the experimental setup, main comparison results, the effectiveness of Cluster-Balanced K', strategy stability analysis, and LLM agent behavior.

A. Dataset and Experimental Protocol

We conduct experiments on the Landslide4Sense benchmark dataset [2], which is specifically designed for landslide detection research. The dataset contains multi-spectral Sentinel-2 imagery with pixel-level binary annotations (landslide vs. non-landslide). The class imbalance is severe: landslide pixels constitute less than 2.4% of the total image area, yielding an approximately 42:1 background-to-landslide ratio.

Table I summarizes the dataset configuration and active learning protocol. Starting from a 5% initial labeled set, each round queries 88 samples from the unlabeled pool for oracle annotation. After 15 rounds, the labeled set grows to 1,471 samples, covering approximately 48% of the full training pool.

B. Implementation Details

Segmentation Model. We use DeepLabV3+ with a ResNet-50 backbone pre-trained on ImageNet. The model is fine-tuned on the Landslide4Sense dataset at each active learning round.

Training Parameters. Each round trains for 10 epochs with batch size 4, learning rate 1×10^{-4} (Adam optimizer), and standard data augmentation including random horizontal/vertical flips, rotation, and brightness adjustment.

LLM Agent Configuration. The LLM agent is executed through an OpenAI-compatible API endpoint. In the full-model configuration (Variant A), the agent performs candidate inspection and selection finalization, while λ_t is determined by the closed-loop policy. In Variant B, the agent is additionally granted explicit λ-setting authority.

Evaluation Metrics. We report three metrics:

- *Area under the Learning Curve (ALC)*: the normalized integral of mIoU across all 15 rounds, measuring whole-curve label efficiency;
- *Final mIoU*: mean Intersection over Union at the last round, measuring terminal segmentation accuracy;
- *Final F1-Score*: harmonic mean of precision and recall at the last round.

All metrics are reported as 3-seed mean $\pm$ standard deviation.

C. Baseline Strategies

We compare AAL-SD against six baseline active learning strategies spanning three categories:

Uncertainty-based:

- *Entropy Sampling*: selects samples with the highest average pixel-wise prediction entropy;
- *BALD (Bayesian Active Learning by Disagreement)*: estimates uncertainty through the variance of predictions across multiple MC Dropout forward passes.

Diversity-based:

- *Core-Set Sampling*: selects samples that maximize feature-space coverage using a greedy farthest-first strategy.

Hybrid:

- *DIAL-style*: cluster-based diversity with uncertainty constraints, following the Deep Interactive and Active Learning paradigm;
- *Wang-style*: a two-stage $U \to K$ reranking approach that first filters by uncertainty then diversifies.

Control:

- *Random Sampling*: uniformly random selection from the unlabeled pool.

Additionally, we distinguish two full-model AAL-SD variants: **Model A** (policy-controlled λ), which serves as the primary variant, and **Model B** (agent-controlled λ), where the LLM agent sets λ_t directly at each round.

D. Main Results: 15-Round Full AL Cycle

Table II presents the final mIoU after 15 rounds of active learning, averaged across three independent seeds. Several key findings emerge.

First, all directed sampling strategies outperform Random by approximately 0.8–1.6% in final mIoU, confirming the value of informed sample selection even under severe class imbalance. Second, Core-Set achieves the highest final mIoU (0.7101) with remarkably low cross-seed variance (range 0.17%), followed closely by BALD (0.7080) and Entropy (0.7065). These simple, single-criterion strategies prove highly effective on this task.

Third, the proposed AAL-SD variants (Model A: 0.6971; Model B: 0.7027) achieve competitive performance, positioned within the cluster of directed strategies. Model B outperforms Model A by 0.56%, suggesting that granting the agent direct λ-control can improve terminal accuracy. Notably, Model B also exhibits substantially better stability than Model A (range 0.76% vs. 1.99%).

Fourth, DIAL-style (0.6932), despite being a hybrid approach, performs below Random on this task. This result indicates that hybrid strategies with fixed weighting schemes may not be well-suited to severely imbalanced segmentation scenarios where the diversity component's contribution is constrained.

The performance gap between the best baseline (Core-Set, 0.7101) and our primary variant (Model A, 0.6971) is 1.3%. While AAL-SD does not claim state-of-the-art accuracy, it provides a qualitatively different value proposition: interpretable, auditable selection decisions with natural language rationale, which we analyze further in Section IV-G.

E. Cluster-Balanced K' Effectiveness

A key contribution of this work is the Cluster-Balanced K' algorithm (Section III-C3), designed to address the dominant-cluster capture problem observed in the original $K(x)$ metric.

Before K': Cluster Capture. Analysis of the original $K(x)$-based sampling reveals severe cluster capture bias. In the feature-space clustering of the unlabeled pool, Cluster 19 alone accounts for over 52% of all labeled samples, while several minority clusters receive near-zero representation. This occurs because $K(x)$ measures raw distance to the labeled set without accounting for per-cluster saturation: once a cluster accumulates many labeled samples, it paradoxically generates *higher* distances (to the nearest labeled neighbor in other clusters), sustaining a self-reinforcing selection loop.

After K': Balanced Coverage. With the saturation-weighted correction in $K'(x)$, the sampling from Cluster 19 drops to near 0%, and the annotation budget is redistributed to previously underrepresented clusters. The K' correction effectively orthogonalizes the knowledge-gain direction from the uncertainty direction, ensuring that the two signals in the acquisition function provide complementary rather than redundant information.

Lambda Sweep with K'. To verify that K' does not introduce unexpected sensitivity to λ, we conduct a sweep across $\lambda \in \{0.0, 0.1, 0.2, \ldots, 1.0\}$ with the corrected $K'(x)$. The total mIoU range across all λ values is 1.56%, confirming that the flat lambda-sensitivity pattern persists even after the cluster-capture correction. This result separates two phenomena: (1) K' successfully fixes the sampling bias in the knowledge signal, and (2) the limited lambda sensitivity is an intrinsic property of the task rather than an artifact of the original $K(x)$'s dysfunction.

F. Strategy Stability Analysis

Beyond final accuracy, the reliability of an active learning strategy across different random initializations is critical for practical deployment. We measure stability as the range (maximum minus minimum) of final mIoU across three seeds.

The stability ranking reveals a clear pattern: simple fixed strategies achieve the lowest cross-seed variance. Core-Set leads with a range of only 0.17%, followed by Entropy (0.71%) and Model B (0.76%). In contrast, complex or adaptive methods show wider spreads: Model A (1.99%), BALD (2.52%), Random (2.95%), and DIAL (3.27%).

This finding carries an important practical implication. For deployment scenarios where consistency across runs matters more than marginal accuracy gains, simpler fixed strategies such as Core-Set or Entropy offer the most reliable performance. The higher variance of adaptive methods like Model A suggests that the policy's interaction with different random initializations introduces additional stochasticity, though Model B mitigates this substantially through the agent's contextual λ decisions.

G. LLM Agent Analysis

The LLM agent in AAL-SD provides a qualitative capability absent from purely numerical baselines: natural language rationale for each selection decision. We analyze the agent's behavior across the 15-round cycle.

Agent Reasoning Examples. The agent's decision process involves querying system status, inspecting candidate samples, and providing a justification. Representative excerpts include:

> *"Round 5: The overfit risk is moderate (0.42) and the gradient cosine is slightly negative (-0.18). I will prioritize candidates with high uncertainty scores to strengthen boundary learning before shifting toward coverage."* — Model A, Seed 42

> *"Round 11: Performance has plateaued at mIoU 0.748. The gradient cosine has turned strongly negative (-0.58), indicating train-validation divergence. I recommend conservative selection focusing on uncertain samples near decision boundaries."* — Model B, Seed 42

These rationales provide auditable records of the selection logic, a property that is particularly valuable in safety-critical remote sensing applications where human operators need to understand and validate automated decisions.

Model A vs. Model B. The two full-model variants differ in the agent's control authority. In Model A, the agent inspects candidates and finalizes selection, but λ_t is set by the deterministic policy. In Model B, the agent additionally determines

TABLE III: Component Ablation Results (Seed 42)

Configuration	ALC	Final mIoU	Final F1
AAL-SD (full, A)	0.6693	0.7651	0.8490
w/o Agent (score-based)	0.6638	0.7573	0.8424
Fixed $\lambda = 0.5$	0.6669	0.7566	0.8430
Uncertainty-only ($\lambda = 0$)	0.6687	0.7555	0.8409
Knowledge-only ($\lambda = 1$)	0.6601	0.7512	0.8368

λ_t based on its contextual assessment. The results show that Model B achieves higher final mIoU (0.7027 vs. 0.6971) and better stability (range 0.76% vs. 1.99%), suggesting that the agent's contextual reasoning about the appropriate exploration-exploitation balance can complement the deterministic policy.

Guardrail Activations. The selection guardrails (Section III) are activated in approximately 15% of rounds across all seeds, primarily in the early rounds where the overfit risk is volatile. When activated, the guardrails override the agent's selection to enforce conservative sampling, preventing potentially destabilizing acquisitions. This mechanism ensures that the agent's flexibility does not compromise learning stability.

H. Component Ablation

To isolate the contribution of each component, we conduct ablation experiments on a representative seed (Seed 42).

Table III presents the ablation results. Removing the LLM agent and falling back to pure score-based selection (argmax) reduces ALC from 0.6693 to 0.6638 (0.55% decrease), demonstrating that the agent's reasoning provides improvements beyond numerical ranking. Fixing λ at 0.5 instead of using the adaptive policy reduces both ALC and final mIoU, validating the value of dynamic weighting. The comparison between uncertainty-only ($\lambda = 0$, ALC=0.6687) and knowledge-only ($\lambda = 1$, ALC=0.6601) configurations confirms that both signals contribute, with uncertainty being more critical in the early learning stages.

V. Discussion

A. Practical Implications for Remote Sensing Practitioners

The experimental results carry direct implications for practitioners seeking to deploy active learning for landslide segmentation or similar imbalanced remote sensing tasks.

For scenarios where annotation budget is limited and maximizing segmentation accuracy is the primary objective, our results suggest that simple directed strategies offer an excellent cost-performance trade-off. Core-Set sampling achieves the highest final mIoU (0.7101) with the lowest cross-seed variance (0.17%), requiring no hyperparameter tuning beyond the standard feature extractor. Entropy sampling provides a close alternative (0.7065) with minimal implementation complexity. These findings echo recent observations in the broader active learning literature that sophisticated acquisition functions do not consistently outperform simpler baselines on real-world tasks.

However, accuracy alone does not capture the full value of an active learning system. In operational settings such as disaster response mapping or geological survey automation, the ability to *explain* why specific samples were selected for annotation is essential for building trust with domain experts and meeting audit requirements. AAL-SD addresses this need through the LLM agent's natural language rationale, which documents the reasoning behind each acquisition decision. This interpretability comes at a modest accuracy trade-off (1.3% below Core-Set) but provides a qualitatively different capability that numerical baselines cannot offer.

B. Lambda Sensitivity: An Unexpected Observation

Interestingly, our experiments reveal that the performance difference across λ values remains within 2% in all tested conditions, suggesting that the knowledge-uncertainty balance may be less critical than commonly assumed for severely imbalanced segmentation tasks. This observation holds both before and after the K' correction (lambda sweep range of 1.56% with K'), and the cross-λ variation is comparable to or smaller than the cross-seed variation for most strategies.

One plausible explanation is that under severe class imbalance (42:1 in Landslide4Sense), the minority-class samples that are most task-critical occupy a spatially compact and relatively uniform region in the feature space. Consequently, both uncertainty-driven and diversity-driven selection converge on similar subsets of informative samples, regardless of their relative weighting. In other words, the "task-critical statistics" of the selected batches may be approximately invariant to the acquisition criterion when the minority class constrains the effective search space.

A detailed investigation of this phenomenon, including its theoretical underpinnings and cross-task validation, is beyond the scope of this system-oriented paper and will be addressed in future work. Nevertheless, this observation has a practical consequence: for practitioners using AD-KUCS, the choice of λ is not a sensitive hyperparameter, and the default three-phase policy provides a safe, reasonable operating point without requiring task-specific tuning.

C. Limitations

Several limitations of this work should be acknowledged. First, all experiments are conducted on a single dataset (Landslide4Sense) with a specific type of class imbalance. The generalizability of our findings to other remote sensing tasks (e.g., urban mapping, crop classification) or different imbalance ratios remains to be validated.

Second, the LLM agent introduces additional computational cost and latency. Each round requires multiple API calls for system status queries, candidate inspection, and selection finalization. While this overhead is negligible compared to the annotation cost in practical workflows, it adds complexity to the deployment pipeline and introduces a dependency on external API availability.

Third, the lambda sensitivity observation, while consistent across all our experimental conditions, may not hold for balanced or mildly imbalanced datasets where the diversity signal plays a more discriminative role. Our results should therefore be interpreted in the context of severe class imbalance rather than as a general statement about active learning.

Finally, the current evaluation uses a fixed segmentation architecture (DeepLabV3+ with ResNet-50). The interaction between the active learning strategy and model architecture could affect the relative ranking of methods, particularly for architectures with different feature space geometries.

VI. Conclusion

This paper presented AAL-SD, an LLM-agent-driven adaptive active learning framework for landslide semantic segmentation from remote sensing imagery. The framework introduces a dual-line architecture that separates the data acquisition flow (Active Learning Line) from the optimization diagnostic flow (Learning Control Line), enabling gradient-derived risk signals to regulate the acquisition policy in a closed-loop manner.

The core acquisition module, AD-KUCS, fuses uncertainty and knowledge gain through an adaptive weighting mechanism governed by a three-phase lambda policy. To address the dominant-cluster capture problem inherent in the original knowledge gain metric $K(x)$, we proposed Cluster-Balanced $K'(x)$, which applies saturation-weighted corrections to redistribute the annotation budget across underrepresented feature clusters. The LLM agent provides interpretable candidate inspection and selection finalization with natural language rationale, offering an auditable decision-making interface for safety-critical remote sensing applications.

Comprehensive evaluation on the Landslide4Sense benchmark with 15 active learning rounds across three independent seeds demonstrates that AAL-SD achieves competitive performance against established baselines including Core-Set, Entropy, and BALD sampling. While simple fixed strategies such as Core-Set attain the highest accuracy and stability, AAL-SD uniquely combines adaptive control, cluster-balanced sampling, and interpretable agent reasoning in a unified framework. Component ablation confirms the positive contribution of both the LLM agent and the dynamic weighting mechanism.

An unexpected but consistent observation across all experiments is that lambda sensitivity remains within 2%, comparable to cross-seed variance. This suggests that under severe class imbalance, the knowledge-uncertainty balance is a safe rather than sensitive control variable, favoring robust deployment over fine-tuned optimization.

Future work will pursue three directions: (1) cross-task validation on balanced and mildly imbalanced datasets to test the generalizability of the lambda insensitivity observation; (2) theoretical analysis of the conditions under which acquisition strategy equivalence emerges; and (3) integration of more advanced segmentation architectures to evaluate the framework's architecture-agnostic potential.

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